Generation Depth Fingerprinting via Differential Depth Response in Non-Paraphrastic LLM Chains

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Abstract

Web-scraped training corpora increasingly contain LLM-generated text of unknown provenance depth. Current detection methods offer only binary real-vs-synthetic classification, conflating qualitatively different material: first-generation text retains much of the source distribution's lexical properties, while deeper recursive generations exhibit measurably reduced vocabulary diversity and compressed perplexity. We formulate **generation depth classification** (gen-0 / gen-1 / gen-2+) as a new task beyond binary detection and characterize **Differential Depth Response** (DDR): central-tendency features (entropy mean, type–token ratio, hapax ratio) drift monotonically with depth, while shape features (entropy max, tail decay) remain near-stationary—consistent with recursive generation inducing a location shift in the NLL distribution while approximately preserving distributional shape. On 8 base-model generators from 5 families (0.6B–8B), a lightweight 7-feature ensemble achieves leak-free binary $F_1 = 0.694$ under leave-one-generator-out (LOGO) evaluation and 0.830 on cross-model chains, using a single 410M reference model. Causality analysis confirms DDR with 6/7 Granger-significant features and a $5.3\times$ permutation importance gap between drifting and stationary features. Depth-aware data curation is statistically equivalent to random subsampling for downstream causal language model quality (TOST $p = 1.1\times10^{-4}$ at $\pm2\%$ margin), establishing depth fingerprinting as a *diagnostic* tool for provenance characterization, not a curative tool for data selection. The signal degrades substantially under instruction tuning, restricting current scope to base models.

1 Introduction

Large language models (LLMs) now generate text that is increasingly difficult to distinguish from human-written content. As LLM-generated text proliferates on the web—estimates suggest 7–17% of recent scientific papers contain substantial AI-generated content—it inevitably re-enters training corpora for subsequent models. This creates recursive generation chains: a human writes a seed text (generation depth 0), an LLM produces a continuation (depth 1), another LLM may further elaborate on that output (depth 2), and so on. Understanding where a given text sits along this chain—its *generation depth*—is crucial for data governance, training corpus curation, and understanding the feedback loops that drive model collapse.

Existing AI-text detectors frame the problem as binary classification: human vs. machine. Methods such as DetectGPT [Mitchell et al., 2023], FastDetectGPT [Bao et al., 2024], Binoculars [Hans et al., 2024], and LD-Score [Tulchinskii et al., 2024] achieve strong binary discrimination but cannot distinguish first-generation text from deeper recursive generations. This conflation matters: gen-1 text retains substantial distributional similarity to its human source, while gen-2+ text exhibits measurably compressed vocabulary diversity and perplexity—qualitatively different material that may differentially affect downstream model quality. Separately, work on paraphrastic chains [Briesch et al., 2025] has shown attractor convergence behavior limiting depth discriminability to approximately 3 classes, but this depends on semantic-preservation constraints absent in non-paraphrastic continuation chains.

In this paper, we formulate **generation depth classification** as a new task and characterize the distributional signatures that enable it. We introduce the concept of **Differential Depth Response** (DDR): when text is recursively generated through non-paraphrastic chains, central-tendency and diversity features of the token-level negative log-likelihood (NLL) distribution (entropy mean, type–token ratio, hapax ratio) *drift* monotonically with depth, while shape features (entropy max, tail decay) remain *near-*

stationary. This is consistent with recursive generation inducing a location shift in the NLL distribution while approximately preserving distributional shape.

Our contributions are:

1. **Task formulation.** We define generation depth classification (gen-0/gen-1/gen-2+) as a tractable ordinal classification problem beyond binary detection. No prior work addresses depth estimation in non-paraphrastic continuation chains.
2. **Empirical characterization of DDR.** We validate DDR across 8 base-model generators from 5 model families (Pythia, OLMo, Qwen, Llama, Mistral; 0.6B–8B scale), 3 prompt types, and 4 text domains. A 7-feature ensemble achieves leak-free binary $F_1 = 0.694$ under LOGO and 0.830 on cross-model chains. Causality analysis (6/7 Granger-significant features, $5.3\times$ permutation importance gap) confirms DDR as a genuine empirical phenomenon beyond vocabulary impoverishment.
3. **Negative finding on downstream utility.** Depth-aware data curation does not improve downstream CLM quality compared to random subsampling (TOST equivalence confirmed at $\pm 2\%$ margin). Depth fingerprinting is a diagnostic tool for provenance characterization, not a curative tool for data selection.

We transparently report scope limitations: the signal degrades substantially under instruction tuning and high sampling temperatures ($t > 0.9$), and 5-class ordinal resolution is not achievable with current features (ceiling at 3 classes).

2 Related Work

Perplexity-based AI-text detection. Statistical detection methods leverage the observation that LLM-generated text occupies lower-perplexity regions of the model's output distribution. DetectGPT [Mitchell et al., 2023] uses perturbation-based log-probability curvature; FastDetectGPT [Bao et al., 2024] replaces perturbations with conditional probability sampling for efficiency. Binoculars [Hans et al., 2024] compares cross-entropy ratios between two models. GLTR [Gehrmann et al., 2019] visualizes token-level statistics. DNA-GPT [Yang et al., 2024] employs divergent continuation comparison. All operate as binary classifiers; none address ordinal depth estimation.

Distributional signatures and LD-Score. Tulchinskii et al. [2024] propose character-level distributional signatures (LD-Score) for authorship verification and AI detection, demonstrating that distributional divergence at the character level captures stylistic fingerprints. We incorporate LD-Score as both a baseline and an augmentation feature (Appendix E).

Model collapse and recursive generation. Theoretical work on model collapse [Shumailov et al., 2024] predicts that recursive training on synthetic data leads to distributional narrowing and eventual mode collapse. Dohmatob et al. [2024] formalize the rate of distributional degradation. Our work complements this line by providing an empirical method to *detect* the depth signature of recursive generation in individual texts, rather than studying the aggregate distributional consequences for model training.

Iterative generation chains. Briesch et al. [2025] study paraphrastic LLM chains and find attractor convergence limiting depth discriminability to approximately 3 semantic clusters. Our work studies non-paraphrastic chains (continuation, elaboration), where the absence of semantic-preservation constraints allows distributional drift to persist to deeper levels.

Vocabulary diversity and stylometry. Type–token ratio (TTR), hapax legomena ratio, and entropy-based measures have a long history in computational stylometry and authorship attribution [Tweedie and Baayen, 1998]. We repurpose these as depth-discriminative features and show they capture the "drifting" component of DDR.

Data quality and curation. Perplexity-based filtering [Wenzek et al., 2019] and quality classifiers are standard in training data pipelines. Our Claim 3 (negative finding) directly tests whether depth information adds value to such curation, finding it does not.

3 Method

3.1 Problem Formulation

Given a text x, we aim to predict its *generation depth* $d \in \{0, 1, 2+\}$, where $d = 0$ denotes human-written seed text, $d = 1$ denotes first-generation LLM output from a human seed, and $d = 2+$ denotes text produced at depth ≥ 2 in a recursive generation chain. The primary evaluation metric is **leak-free binary $\mathbf{F_1}$**: the F_1 for discriminating gen-1 from gen-2+, excluding gen-0 samples to avoid inflated performance from seed-sharing cues.

3.2 Chain Construction

We construct recursive generation chains of depth 0–4 using base-model LLMs. Given a human-written seed text x_0 (depth 0), we prompt a generator G to produce $x_1 = G(x_0)$ (depth 1), then $x_2 = G(x_1)$ (depth 2), and so on up to x_4. We evaluate three prompt types: *continuation* (unconstrained generation), *elaboration* (expand on the input), and *inspired rewriting* (write new text inspired by the input). Seed texts are drawn from four domains: Wikipedia, news, academic, and code.

3.3 Feature Extraction

We extract 7 features from each text using a reference language model (Pythia-410M by default). For each token t_i in the text, we compute its negative log-likelihood (NLL) under the reference model, yielding an NLL sequence. The 7 features are:

- **Surface features** (2): type–token ratio (TTR) and hapax legomena ratio (hapax_ratio)—lexical diversity measures computed directly from the token sequence.
- **Reference features** (5): entropy mean, entropy standard deviation, entropy minimum, entropy maximum, and tail decay—summary statistics of the NLL distribution extracted via the reference model.

3.4 Differential Depth Response (DDR)

We observe that features partition into two behavioral categories as generation depth increases:

- **Drifting features**: hapax ratio ($\rho \approx -0.63$), TTR ($\rho \approx -0.61$), and entropy mean ($\rho \approx -0.60$) exhibit strong monotonic decrease with depth.
- **Stationary features**: tail decay ($\rho \approx 0.04$), entropy max ($\rho \approx 0.05$), and entropy std ($\rho \approx -0.12$) remain approximately constant.

This differential response is consistent with recursive generation inducing a *location shift* in the NLL distribution: central-tendency indicators track this shift and therefore drift, while shape indicators reflect distributional form and remain stable.

3.5 Classification

We train an XGBoost gradient-boosted ensemble on the 7-feature representation. For evaluation, we use leave-one-generator-out (LOGO) cross-validation: for each fold, one generator's data is held out as the test set and the remaining generators form the training set. This ensures no information about the test generator leaks into training. We additionally report leave-one-family-out (LOFO) results, where all generators from the same model family are held out simultaneously.

Table 1: Feature ablation (5-fold CV, Pythia-2.8B).

Feature Set	**N**	**Macro F_1**	**95% CI**
Full	7	0.820	[0.807, 0.833]
Surface-only	2	0.763	[0.747, 0.774]
Reference-only	5	0.719	[0.706, 0.733]

Table 2: Feature–depth Spearman correlations (Pythia-2.8B).

Feature	ρ	**Category**	**Trend**
hapax_ratio	-0.63	Drifting	Monotonic decrease
TTR	-0.61	Drifting	Monotonic decrease
entropy_mean	-0.60	Drifting	Monotonic decrease
entropy_min	-0.39	Intermediate	Moderate decrease
entropy_std	-0.12	Stationary	Near-flat
entropy_max	$+0.05$	Stationary	Near-flat
tail_decay	$+0.04$	Stationary	Near-flat

4 Experiments

4.1 Feature Ablation and DDR Characterization

We first characterize DDR on Pythia-2.8B chains with 5-fold cross-validation. Table 1 shows feature ablation results. The full 7-feature set achieves macro F_1 = 0.820 (95% CI [0.807, 0.833]), significantly outperforming both surface-only (0.763; [0.747, 0.774]) and reference-only (0.719; [0.706, 0.733]) subsets. Surface features capture stronger individual signals, but reference features provide complementary information (McNemar $p \approx 0$ for full vs. each subset), confirming feature complementarity rather than redundancy.

Table 2 quantifies DDR with Spearman correlations between each feature and generation depth. Drifting features show $|\rho| > 0.60$ with monotonic decrease, while stationary features show $|\rho| < 0.12$. Perplexity decreases monotonically with depth (median: 13.50 $\rightarrow$ 4.96 $\rightarrow$ 3.00 $\rightarrow$ 2.07 $\rightarrow$ 1.72), prevalidating that the reference model captures a progressively more "expected" signal at deeper generations.

4.2 Cross-Generator Generalization

We evaluate on 5 generators (Pythia-2.8B, Pythia-1B, OLMo-2-1B, Qwen3-1.7B, Qwen3-0.6B; 18,423 total records) under LOGO cross-validation. Table 3 reports per-generator results. The average LOGO macro F_1 is 0.804 and the average binary F_1 is 0.694, with Pythia-1B held-out highest (binary 0.724) and OLMo-2-1B lowest (binary 0.663). The family-out (LOFO) delta is only 0.011 pp, confirming cross-family generalization. Feature importance ranks hapax_ratio (0.289) $>$ TTR (0.168) $>$ entropy_mean (0.160), consistent with surface drifting features as the primary drivers.

4.3 Baseline Comparison

We compare the 7-feature ensemble against existing detection methods repurposed as ordinal depth predictors (Table 4). The 7-feature ensemble (LOGO F_1 = 0.798; 95% CI [0.786, 0.799]) significantly outperforms all single-detector baselines (bootstrap $p = 0$ for all pairwise comparisons). The strongest baseline, Avg-LogProb (F_1 = 0.765), confirms that ordinal depth signal is partially capturable by simple log-probability statistics. However, the 7-feature ensemble achieves a narrower confidence interval,

Table 3: LOGO cross-generator evaluation (5 generators).

Held-out Generator	**Macro F_1**	**Binary F_1**	**Gen-1 F_1**
Pythia-1B	0.851	0.724	0.689
Qwen3-0.6B	0.824	0.710	0.652
Qwen3-1.7B	0.805	0.698	0.641
Pythia-2.8B	0.804	0.694	0.638
OLMo-2-1B	0.752	0.663	0.585
Average	**0.804**	**0.694**	**0.641**

Table 4: Ordinal baseline comparison under LOGO.

Method	**LOGO F_1**	**95% CI**	ρ	**Δ vs 7-feat**
7-feature ensemble	0.798	[0.786, 0.799]	—	—
Avg-LogProb	0.765	[0.616, 0.820]	0.788	-0.033
LD-Score	0.721	[0.588, 0.770]	0.794	-0.077
FastDetectGPT	0.627	[0.511, 0.762]	0.433	-0.171
Binoculars	0.579	[0.405, 0.664]	0.199	-0.219

reflecting more stable cross-generator generalization. FastDetectGPT (0.627) and Binoculars (0.579) perform poorly as ordinal predictors because they were designed for binary discrimination.

4.4 Cross-Model Chains and Cross-Reference Validation

Cross-model chains. Real-world contamination likely involves heterogeneous generators at each depth step. We construct cross-model chains where different generators produce each successive depth level. Table 5 shows that cross-model chains yield *higher* binary F_1 (0.830) than single-model chains (0.694), a $+13.6$ pp improvement. Generator heterogeneity amplifies distributional discontinuities between depth levels, making depth more discriminable.

Cross-reference model validation. A key concern is whether DDR is an artifact of the Pythia-410M reference model. We extract features using three different reference models (Table 6). All three produce negative entropy drift and comparable LOGO F_1 (0.803–0.816, $<$1.5 pp range), confirming that DDR is a generation-intrinsic phenomenon.

4.5 DDR Causality Analysis

To establish DDR as more than correlational, we conduct three causality tests (Table 7).

Permutation importance. Drifting features have 5.3$\times$ higher average importance (0.158) than stationary features (0.030), confirming their dominant role in classification.

Granger causality. Panel Granger tests ($n = 29{,}095$) show 6/7 features are Granger-significant for depth prediction. The single non-significant feature is entropy_min ($p = 0.12$), consistent with its intermediate status.

Feature intervention. Single-feature interventions produce a maximum F_1 drop of only 0.020, indicating that the depth signal is distributed across the drifting feature group rather than concentrated in any single feature.

Table 5: Cross-model vs. single-model chain evaluation.

Setting	Binary F_1	3-class F_1	N Gen.
Single-model LOGO	0.694	0.762	8 (homogeneous)
Cross-model LOGO	0.830	0.789	5 (heterogeneous)
Δ	$+0.136$	$+0.027$	—

Table 6: Cross-reference-model DDR validation.

Reference Model	LOGO F_1	entropy_mean drift	DDR
Pythia-410M (default)	0.803	$[-1.19, -1.40]$	✓
OLMo-2-1B	0.807	$[-0.61, -0.91]$	✓
Qwen3-0.6B	0.816	$[-0.96, -1.35]$	✓

4.6 Downstream Utility: A Negative Finding

We test whether depth information can improve downstream training data curation by fine-tuning OLMo-1B on depth-stratified subsets (3 seeds each; Table 8). Size-matched random subsampling (ppl = 2.367) *outperforms* depth 0–1 filtering (2.387) and binary filtering (2.389). TOST equivalence testing confirms statistical equivalence at $\pm 2\%$ margin ($p = 1.1 \times 10^{-4}$). Only the depth-0-only variant (ppl = 2.681) shows meaningful degradation, attributable to the small sample size (797 vs. 3,951). Across seeds, variance is negligible ($<$0.002 ppl), strengthening the conclusion.

This negative finding terminates a seemingly plausible research direction: although depth information is detectable, it has no curative value for data selection. Depth fingerprinting is positioned as a **diagnostic** tool for provenance characterization.

4.7 Instruct Model Degradation

The DDR framework is developed and evaluated on base models. We systematically test three instruct-tuned models (Table 9). Performance degrades substantially: Qwen3-1.7B LOGO binary F_1 drops from 0.821 (base) to 0.658 (instruct), a -16.3 pp gap. OLMo-2-1B instruct shows 6/7 features with Spearman direction reversal, and Llama-3.2-3B instruct reverses 3/7 features. Instruction tuning imposes length regulation and format constraints that compress the distributional variation DDR exploits. This is the primary scope limitation: our method currently applies to base models only.

4.8 Multi-Domain Evaluation

To assess ecological validity, we evaluate on seeds from 4 domains (500 seeds/domain $\times$ 5 depths; Table 10). All non-Wikipedia domains achieve binary F_1 in the range 0.739–0.754, with a maximum inter-domain gap of only 1.5 pp (excluding the Wikipedia baseline at 0.815). Academic text is hardest (0.739) due to higher intrinsic vocabulary diversity; code is easiest (0.754) due to bimodal entropy distributions that amplify DDR. All confidence intervals exclude 0.5, confirming significant depth signal across all domains.

4.9 Large-Scale Generator

We add Qwen3-8B-Base to test DDR at frontier scale (Table 11). With 6 generators, the average LOGO F_1 is 0.789 (-1.1 pp vs. 5-gen baseline of 0.800). Qwen3-8B held-out F_1 is 0.683—lower than average but still well above chance. Higher-capacity models compress feature margins (hapax $\rho = -0.558$ vs. -0.63 for smaller models) but the DDR pattern (drifting vs. stationary) is fully preserved.

Table 7: DDR causality analysis.

Feature	Perm. Imp.	Granger sig.	Category
hapax_ratio	0.289	✓ ($p < 0.001$)	Drifting
TTR	0.168	✓ ($p < 0.001$)	Drifting
entropy_mean	0.160	✓ ($p < 0.001$)	Drifting
entropy_max	0.063	✓ ($p < 0.01$)	Stationary
entropy_std	0.046	✓ ($p < 0.05$)	Stationary
entropy_min	0.015	× ($p = 0.12$)	Stationary
tail_decay	0.012	✓ ($p < 0.05$)	Stationary

Table 8: Downstream CLM perplexity by training data composition (3 seeds, OLMo-1B).

Variant	N train	Ppl (mean ± std)	Δ vs size-matched
All data	12,299	2.335 ± 0.0003	−0.032
Size-matched random	3,951	2.367 ± 0.0002	— (baseline)
Depth 0+1 only	3,951	2.387 ± 0.0005	+0.020
Binary-filtered	3,951	2.389 ± 0.0004	+0.022
Depth 0 only	797	2.681 ± 0.0012	+0.314

5 Discussion

DDR as location shift. The differential behavior of drifting vs. stationary features is consistent with a location-shift model: recursive generation shifts the NLL distribution's central tendency (lower perplexity, reduced diversity) while approximately preserving its shape (tail behavior, maximum entropy). This is not a new theoretical contribution—it follows from known properties of recursive distributional transformation—but the empirical validation across 8 generators, 5 families, 4 domains, and 3 prompt types is novel.

Practical implications. The cross-model chain result (binary $F_1 = 0.830$) is encouraging for real-world deployment, where contamination chains likely involve heterogeneous generators. The 7-feature ensemble requires only a single 410M reference model forward pass, making it substantially cheaper than QLoRA-based classifiers ($\sim$15$\times$ inference cost advantage; Appendix F).

Limitations. (1) **Base models only**: instruct-tuned models show -16.3 pp degradation (Qwen) with feature direction reversal. (2) **Temperature sensitivity**: signal degrades at $t > 0.9$ (Appendix C). (3) **3-class resolution ceiling**: 5-class F_1 drops to $\sim$0.50; ordinal methods offer no improvement (Appendix B). (4) **Scale effect**: larger generators (8B) show moderate degradation (held-out $F_1 = 0.683$). (5) **Prompt-type dependence**: continuation chains show stronger drift ($F_1 = 0.781$) than elaboration (0.730; Appendix I).

6 Conclusion

We formulated generation depth classification as a new task beyond binary AI-text detection and characterized Differential Depth Response (DDR) in non-paraphrastic LLM generation chains. A lightweight 7-feature ensemble achieves leak-free binary $F_1 = 0.694$ under 8-generator LOGO evaluation (per-generator range: 0.633–0.724) and 0.830 on cross-model chains. DDR is validated as a genuine empirical phenomenon through causality analysis (6/7 Granger-significant features, 5.3$\times$ permutation importance gap) and cross-reference model validation (3 reference models, $<$1.5 pp variance). Depth-aware data

Table 9: Instruct vs. base model DDR comparison.

Model	Type	LOGO Bin. F_1	3-class F_1	Feat. Rev.
Qwen3-1.7B	Base	0.821	0.769	0
Qwen3-1.7B	Instruct	0.658	0.543	ρ direction
OLMo-2-1B	Base	0.845	0.754	0
OLMo-2-1B	Instruct	0.813	0.575	6/7
Llama-3.2-3B	Base	0.849	0.742	0
Llama-3.2-3B	Instruct	0.838	0.544	3/7

Table 10: Multi-domain evaluation.

Domain	Binary F_1	95% CI	3-class F_1
Wikipedia (baseline)	0.815	[0.799, 0.831]	0.863
Code	0.754	[0.737, 0.770]	0.634
News	0.746	[0.726, 0.764]	0.652
Academic	0.739	[0.720, 0.757]	0.622

curation is statistically equivalent to random subsampling for downstream CLM quality, establishing depth fingerprinting as a diagnostic tool for provenance characterization. The signal degrades under instruction tuning and high sampling temperatures, defining clear scope boundaries for future work on extending DDR to broader generation settings.

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Table 11: 6-generator LOGO with Qwen3-8B.

Generator	**LOGO F_1**	**hapax ρ**	**Scale**
Pythia-1B	0.856	-0.65	1.0B
Qwen3-0.6B	0.824	-0.63	0.6B
Qwen3-1.7B	0.805	-0.61	1.7B
Pythia-2.8B	0.804	-0.63	2.8B
OLMo-2-1B	0.764	-0.60	1.5B
Qwen3-8B	0.683	-0.56	8.0B

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A Multi-Detector Ensemble

We compare the 7-feature ensemble against a multi-detector ensemble combining 4 detectors (FastDetectGPT, Binoculars, LD-Score, Avg-LogProb) plus surface features (Table 12). The multi-detector ensemble achieves a marginal improvement of $+2.1$ pp (3-class) and $+0.3$ pp (binary) at $2.4\times$ inference cost and $3.2\times$ GPU memory. The binary F_1 gap (0.3 pp) is essentially negligible, confirming that the gen-1/gen-2+ boundary is an intrinsic bottleneck that additional features cannot overcome.

Table 12: Multi-detector ensemble comparison.

Model	**3-class F_1**	**Binary F_1**	**Cost Ratio**
7-feature ensemble	0.774	0.705	$1.0\times$
4-det + surface	0.795	0.708	$2.4\times$

B Finer-Grained Ordinal Classification

We test 5-class ordinal classification (gen-0 through gen-4; Table 13). All methods—standard XGBoost, CORAL, and Frank-Hall—achieve only $\sim$0.50 F_1, a -26 pp drop from 3-class. The bottleneck is the gen-1/gen-2 boundary, which concentrates 86% of binary false negatives. Gen-3 vs. gen-4 is nearly indistinguishable ($F_1 \approx 0.35$). Ordinal methods provide no improvement over standard classification, confirming that the resolution ceiling is at 3 classes—a feature-space limitation, not a methodological one.

Table 13: Ordinal classification results.

Method	**3-class F_1**	**5-class F_1**	Δ
XGBoost (standard)	0.762	0.500	-0.262
CORAL	0.758	0.490	-0.268
Frank-Hall	0.760	0.500	-0.260

C Temperature Sensitivity

Table 14 shows that DDR is reliable within $t \in [0.5, 0.7]$ but degrades substantially at $t = 0.9$. The entropy_mean Spearman ρ drops from -0.600 (at $t = 0.7$) to -0.302 (at $t = 0.9$): high-temperature sampling noise drowns out the deterministic depth-wise drift component.

Table 14: Temperature sensitivity.

Temperature	**Within-CV Bin. F_1**	**LOGO Bin. F_1**	**entropy_mean ρ**
0.5	0.843	0.814	-0.584
0.7 (default)	0.947	0.694	-0.600
0.9	0.582	0.480	-0.302

D Human Edit Robustness

Table 15 shows graceful degradation under simulated human editing. At 5% random word replacement, binary F_1 drops only 4.6 pp. LLM rephrasing (average 1.69 sentences replaced, 50.7% edit distance) pro-

duces a comparable drop (-4.5 pp), suggesting that LLM editing does not introduce out-of-distribution perturbations in feature space. Even at 20% replacement, performance remains above random (F_1 = 0.591).

Table 15: Robustness to text editing.

Condition	**Binary F_1**	**Δ vs baseline**
No editing (baseline)	0.695	—
5% random replacement	0.650	-0.046
10% random replacement	0.614	-0.081
20% random replacement	0.591	-0.104
LLM rephrase (avg 1.69 sent)	0.647	-0.045

E LD-Score Augmentation

Adding LD-Score as an 8th feature yields a marginal improvement of $+0.5$ pp (Table 16). LD-Score alone achieves LOGO F_1 = 0.721 (Spearman $\rho = -0.794$ with depth), confirming character-level information partially captures depth signal but is highly correlated with existing reference features. We retain the 7-feature set as the primary model for simplicity.

Table 16: LD-Score augmentation.

Feature Set	**LOGO F_1**	**Δ**
7-feature (default)	0.798	—
7-feature + LD-Score	0.803	$+0.005$
LD-Score only	0.721	-0.077

F End-to-End QLoRA Classifier

Table 17 compares QLoRA fine-tuned models against the 7-feature ensemble. QLoRA models achieve 3-class F_1 of 0.859–0.907, surpassing the 7-feature ensemble by 10–15 pp. However, the 5-class ceiling persists ($F_1 \approx 0.60$), confirming that the resolution limit is task-intrinsic rather than feature-engineering-specific. QLoRA inference costs $\sim$15$\times$ more than the 7-feature pipeline (which requires only a single 410M forward pass), positioning the 7-feature ensemble on efficiency advantage.

Table 17: End-to-end QLoRA vs. 7-feature ensemble.

Model	**Type**	**3-class F_1**	**Binary F_1**	**5-class F_1**
Qwen3-1.7B (instruct)	QLoRA	0.907	0.867	0.551
OLMo-1B (instruct)	QLoRA	0.888	0.836	0.598
Qwen3-1.7B (base)	QLoRA	0.904	0.860	0.554
OLMo-1B (base)	QLoRA	0.859	0.793	0.540
7-feature ensemble	XGBoost	0.762	0.694	0.500

G Text Length Confound Control

Text length shows weak correlation with depth (Spearman $\rho \in [-0.066, +0.134]$), ruling it out as a primary confound. Length-normalized features (TTR, hapax computed on fixed-length windows) preserve and even strengthen DDR trends. Length-matched evaluation (Table 18) shows a -4.6 pp drop ($0.770 \rightarrow 0.724$), but a random subsample control drops by the same amount, confirming this reflects deconfounding rather than information loss.

Table 18: Length confound analysis.

Condition	**Binary F_1**	**Δ vs unmatched**
Unmatched (full)	0.770	—
Length-matched	0.724	-0.046
Random subsample control	0.724	-0.046

H 9-Generator Scale Evaluation

Extending to 9 generators (adding Gemma-2-2B, Llama-3.2-3B, Mistral-7B-v0.3; 42,015 total records), LOGO 3-class F_1 is 0.751 (-1.9 pp vs. 8-gen; Table 19). Gemma-2-2B is a notable outlier (3-class 0.628, binary 0.440) due to SentencePiece tokenizer mismatch with the BPE-based reference model. Mistral-7B (3-class 0.605) reflects 7B-scale signal compression. Excluding the Gemma anomaly, all generators exceed $F_1 = 0.60$.

Table 19: 9-generator per-generator LOGO F_1.

Generator	**3-class F_1**	**Binary F_1**	**Family**
Pythia-1B	0.857	0.724	Pythia
Qwen3-0.6B	0.819	0.710	Qwen
Pythia-2.8B	0.812	0.700	Pythia
Qwen3-1.7B	0.811	0.698	Qwen
OLMo-2-1B	0.773	0.663	OLMo
Llama-3.2-3B	0.759	0.680	Llama
Qwen3-8B	0.696	0.647	Qwen
Gemma-2-2B	0.628	0.440	Gemma
Mistral-7B-v0.3	0.605	0.633	Mistral

I Prompt Type and Chain Type Comparison

Prompt types. Table 20 compares three prompt types on Pythia-2.8B. Continuation shows the strongest within-type signal (CV $F_1 = 0.781$) but lowest depth retention (83.1%), while elaboration has the highest retention (98.6%) but weakest signal (0.730). This is consistent with the DDR location-shift interpretation: stronger semantic constraints narrow the output distribution, reducing cross-depth variation. Leave-one-prompt-type-out (LOPTO) average F_1 is 0.753, only 0.4 pp below within-type average, indicating high cross-prompt-type transferability.

Paraphrastic vs. non-paraphrastic chains. Table 21 compares continuation and paraphrasing chains. Paraphrasing chains show substantially weaker depth signal (CV $F_1 = 0.62$ vs. 0.781, -16.1 pp) because semantic-preservation constraints induce attractor convergence by depth 2, suppressing the distributional drift that DDR exploits. This validates the task formulation: depth estimation is meaningful specifically for non-paraphrastic chains.

Table 20: Prompt-type comparison (Pythia-2.8B).

Prompt Type	**CV F_1**	**LOPTO F_1**	**Retention**	**hapax ρ**
Continuation	0.781	0.762	0.831	-0.74
Inspired rewriting	0.737	0.733	0.939	-0.80
Elaboration	0.730	0.762	0.986	-0.83

Table 21: Paraphrastic vs. non-paraphrastic chains.

Chain Type	**CV F_1**	**hapax ρ**	**entropy_mean ρ**	**Retention**
Continuation	0.781	-0.74	-0.56	0.831
Paraphrasing	0.620	-0.25	-0.18	0.970
Δ	-0.161	$+0.49$	$+0.38$	$+0.139$

J TOST Bootstrap Stationarity Test

We use Two One-Sided Tests (TOST) with 1,000 bootstrap iterations and equivalence margin $d = 0.2$ to formally test feature stationarity (Table 22). Three features (entropy_max, entropy_min, tail_decay) are confirmed stationary (100% rejection rate, median $|d| < 0.1$). entropy_std is rejected (median $d = 0.458$, 0% rejection rate), classifying it as an intermediate "weak drifter" despite its low Spearman ρ (-0.12). This refines the DDR taxonomy: 3 strong drifters + 3 true stationary + 1 intermediate.

Table 22: TOST bootstrap stationarity test.

Feature	**Median d**	**TOST p**	**Rej. Rate**	**Status**
entropy_max	0.062	0.000	1.000	Stationary ✓
entropy_min	-0.016	0.000	1.000	Stationary ✓
tail_decay	0.088	0.000	1.000	Stationary ✓
entropy_std	0.458	1.000	0.000	Intermediate

K Independent Seed Validation

To rule out seed-sharing leakage, we evaluate with independent seed sets per generator (Table 23). The average binary F_1 (0.680) differs from shared-seed LOGO (0.694) by only 1.4 pp, within bootstrap confidence intervals, confirming that shared seeds do not inflate performance.

L Topic Drift Control

Semantic similarity (cosine) increases monotonically with depth (0.622 $\rightarrow$ 0.683 $\rightarrow$ 0.801 $\rightarrow$ 0.859; Table 24), indicating that deeper generations converge semantically. However, DDR feature–depth correlations remain significant after controlling for semantic similarity, and shape features show near-zero correlation with similarity ($|\rho| < 0.17$), confirming that DDR is not merely a proxy for topic drift.

Table 23: Independent vs. shared seed evaluation.

Condition	**Binary F_1**	Δ
Shared seed LOGO	0.694	—
Independent seed (Pythia→OLMo)	0.695	$+0.001$
Independent seed (OLMo→Pythia)	0.666	-0.028
Independent seed avg	0.680	-0.014

Table 24: Semantic similarity by depth transition.

Transition	**Mean Similarity**	**Std**
depth $0 \rightarrow 1$	0.622	0.165
depth $1 \rightarrow 2$	0.683	0.184
depth $2 \rightarrow 3$	0.801	0.201
depth $3 \rightarrow 4$	0.859	0.190